Final Project Timeline

Detecting Gender Stereotypes in Human and AI-Generated Text · Julia Allos & Chris Kang · Computational Linguistics, Prof. Preum

Deliverables: ACL-format paper + interactive demo · Due Sunday, Nov 16, 2026

Week	Focus	Julia	Chris	Joint / Milestone
Wk 1 Oct 15–21	**Setup & data**	Merge CrowS-Pairs + StereoSet (gender only, drop unrelated, join context); split by pair. EDA: class balance, length by label, word frequencies.	GitHub repo + Overleaf (ACL template). Majority-class baseline. BERT mean-pool + all-mpnet-base-v2 embedding code.	Confirm page limit with Prof. Preum. Decide neutral-class fix.
Wk 2 Oct 22–28	**Classifiers**	TF-IDF + logistic regression; extract top-weighted words. Start generation pipeline (occupations, prompt template, model, temperature).	BERT + LR and mpnet + LR. Per-label accuracy / precision / recall / F1 for all models.	All models trained and scored.
Wk 3 Oct 29–Nov 4	**Error analysis**	Review FP/FN sample; tag likely error causes. Collect LLM generations.	Error breakdown by source dataset; TF-IDF vs. BERT disagreement set. Optional small MLP.	**Nov 4: choose official judge.**
Wk 4 Nov 5–11	**Application & demo**	Finish generations; draft Data + TF-IDF + Generation sections.	Scoring pipeline: run judge, aggregate by occupation, pronoun counts. Draft Baseline + BERT + Scoring sections.	Demo together (Gradio/Streamlit on HF Spaces): textbox → label + confidence.
Wk 5 Nov 12–16	**Write & polish**	Intro, Related Work, Limitations/Ethics.	Results tables/figures, Conclusion, Abstract.	**Nov 12: freeze results.** Proofread, test demo link. **Nov 16: submit.**

Key risk

After the unrelated option is dropped, the classifier only learns stereotypical vs. anti-stereotypical. Neutral LLM text ("The nurse checked the chart") is out of distribution and will still be forced into a label. Decide in Week 1: add a neutral class, use a confidence threshold, or report it as a limitation.

Paper checklist (ACL)

Abstract · Introduction · Related Work · Data · Methods · Results · Error Analysis · LLM Application · Conclusion · **Limitations (required)** · Ethics Statement · References · AI Statement